# INTRODUCTION

There is a popular misconception that science is an impersonal, dispassionate, and thoroughly objective enterprise. Whereas most other human activities are dominated by fashions, fads, and personalities, science is supposed to be constrained by agreed rules of procedure and rigorous tests. It is the results that count, not the people who produce them.

This is, of course, manifest nonsense. Science is a people-driven activity like all human endeavor, and just as subject to fashion and whim. In this case fashion is set not so much by choice of subject matter, but by the way scientists think about the world. Each age adopts its particular approach to scientific problems, usually following the trail blazed by certain dominant figures who both set the agenda and define the best methods to tackle it. Occasionally scientists attain sufficient stature that they become noticed by the general public, and when endowed with outstanding flair a scientist may become an icon for the entire scientific community. In earlier centuries Isaac Newton was an icon. Newton personified the gentleman scientist—well connected, devoutly religious, unhurried, and methodical in his work. His style of doing science set the standard for two hundred years. In the first half of the twentieth century Albert Einstein replaced Newton as the popular scientist icon. Eccentric, dishevelled, Germanic, absent-minded, utterly absorbed in his work, and an archetypal abstract thinker, Einstein changed the way that physics is done by questioning the very concepts that define the subject.

Richard Feynman has become an icon for late twentieth-century physics—the first American to achieve this status. Born in New York in 1918 and educated on the East Coast, he was too late to participate in the Golden Age of physics, which, in the first three decades of this century, transformed our worldview with the twin revolutions of the theory of relativity and quantum mechanics. These sweeping developments laid the foundations of the edifice we now call the New Physics. Feynman started with those foundations and helped build the ground floor of the New Physics. His contributions touched almost every corner of the subject and have had a deep and abiding influence over the way that physicists think about the physical universe.

Feynman was a theoretical physicist par excellence. Newton had been both experimentalist and theorist in equal measure. Einstein was quite simply contemptuous of experiment, preferring to put his faith in pure thought.